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Laboratory Activity 4: Analog Input, PWM, and DAC

I. Task

The laboratory activity involved testing analog input, PWM output, and DAC output using an ESP32. Examples 3–5 were performed separately to observe potentiometer readings, LED brightness through PWM, and DAC output voltage.

II. Example 3 – Read a Potentiometer

Potentiometer Readings:

Potentiometer Position	**Raw ADC Reading**	**Measured Voltage**
Fully Clockwise	0	143mV
~25%	659	688mV
~50%	1744	1583mV
~75%	3453	2864mV
Fully Counterclockwise	4095	3155mV

Observation:

As the potentiometer was rotated counterclockwise, the raw ADC reading and measured voltage increased. The lowest recorded raw value was 0, while the highest was 4095.

III. Example 4 – Adjust LED Brightness with a Potentiometer

Raw Input	Predicted PWM Setting	Approx. Duty Cycle	Observed LED Behavior
0	0	0%	OFF
659	41	16%	Dim
1744	108	42%	Medium brightness

3453	215	84%	Bright
4095	255	100%	Fully bright

Observation

The LED brightness increased as the potentiometer reading increased. At a low PWM setting, the LED was dim or OFF. At a high PWM setting, the LED became brighter.

IV. Example 5 – Measure Changing DAC Output

DAC Voltage Measurements

DAC Setting	Predicted Voltage	Measured Voltage
0	0V	0.09V
64	0.83V	0.89V
128	1.66V	1.70V
192	2.48V	2.50V
255	3.30V	3.28V

Observation

The measured DAC voltage increased as the DAC setting increased. The lowest measured voltage was 0.09 V at a DAC setting of 0, while the highest was 3.28 V at a setting of 255. The measured values were close to the predicted values but were not exactly the same.

V. Comparison of Predicted and Observed Results

The observed results were close to the predicted behavior. In the PWM test, increasing the raw potentiometer input increased the PWM duty setting, which made the LED brighter. In the DAC test, increasing the DAC setting also increased the measured output voltage. Small differences were observed between the predicted and measured DAC voltages, such as 0.00 V predicted versus 0.09 V measured at a DAC setting of 0.

PWM and DAC are not the same type of output. PWM rapidly switches the GPIO output between HIGH and LOW and controls the average output through its duty cycle. In contrast, the DAC produces an analog voltage level based on the selected DAC setting. Therefore, the PWM output should not be reported as a DAC voltage.

The ADC endpoint may saturate when the input reaches the limit of the ADC measurement range. This can cause the raw reading to remain at its minimum or maximum value even when the actual voltage changes slightly. In the experiment, the raw ADC reached 0 and 4095 at the extreme positions of the potentiometer.

VI. Conclusion

The laboratory activity demonstrated analog input, PWM control, and DAC output using the ESP32. The potentiometer successfully produced different ADC readings, and these readings were used to control LED brightness through PWM. The DAC also produced increasing voltage levels as the DAC setting increased. The measured results were generally close to the predicted values.